UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

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ANALYSIS OF THE TWIN ROTOR MULTI-INPUT MULTI-OUTPUT (TRMS) SYSTEM

Using Classical Control and State-Space Techniques

Report:

(EE360L) Control Systems Lab Project Based Learning Report

Submitted By

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Abstract

The Twin Rotor Multi-Input Multi-Output (TRMS) system is a nonlinear and highly coupled dynamic system widely used for control system studies. In this project, a TRMS state-space model was implemented in MATLAB, and its transfer function matrix was derived for classical control analysis. Root locus and transient response analyses were performed under proportional feedback control, while state-space techniques including eigenvalue analysis, controllability and observability tests, the PBH criterion, observability Gramian, Hankel singular values, minimal realization, and null-space analysis were used to investigate the system structure. The results showed that the TRMS model is open-loop stable and completely controllable but contains one unobservable state that can be removed without affecting the input–output behavior. The study demonstrates that the integration of classical control and state-space methods provides a comprehensive understanding of the dynamic and structural characteristics of the TRMS system.

Keywords: Twin Rotor Multi-Input Multi-Output (TRMS), State-Space Model, Transfer Function, Root Locus, Controllability, Observability, Minimal Realization, MATLAB, Classical Control.

1 Introduction

The Twin Rotor Multi-Input Multi-Output (TRMS) system is a laboratory-scale dynamic model designed to simulate the behavior of a helicopter. Owing to its nonlinear characteristics, strong coupling effects, and multiple interacting inputs and outputs, the TRMS is widely used as a benchmark for the analysis and design of control systems.

Modern engineering applications frequently involve Multi-Input Multi-Output (MIMO) systems, whose dynamic behavior requires advanced mathematical models for accurate analysis and controller design. The state-space representation provides a complete description of such systems by relating state variables, inputs, and outputs through first-order differential equations. In addition to describing the internal dynamics of a system, the state-space model can be used to derive transfer functions for classical control analysis.

Among classical control techniques, the root locus method is widely employed to investigate the movement of closed-loop poles as controller gain varies. This approach provides valuable information regarding system stability and transient response characteristics, including rise time, settling time, and overshoot.

In this project, the TRMS state-space model was implemented in MATLAB using the Control System Toolbox. The corresponding transfer function matrix was derived, and the principal transfer functions were selected for classical root locus and proportional feedback analyses. To complement the classical investigation, state-space methods including eigenvalue analysis, controllability and observability tests, the PBH criterion, observability Gramian, Hankel singular values, and minimal realization were employed to examine the structural properties of the system.

The objective of this study is to investigate the dynamic and structural characteristics of the TRMS by integrating classical control techniques with state-space analysis, providing a comprehensive understanding of its behavior and internal dynamics.

2 Objectives

The objectives of this project are:

- To implement the TRMS state-space model in MATLAB.
- To derive the transfer function matrix from the state-space representation.
- To identify and analyze the principal transfer functions of the system.
- To perform classical root locus analysis for the selected transfer functions.
- To evaluate the transient response characteristics, including rise time, settling time, peak time, and overshoot.
- To investigate the open-loop stability of the TRMS model through eigenvalue analysis.
- To determine the controllability and observability of the system.
- To study the structural properties of the TRMS model using the PBH criterion, observability Gramian, Hankel singular values, and minimal realization.
- To interpret the dynamic and structural behavior of the TRMS system using classical control and state-space techniques.

3 Methodology

The methodology adopted in this project involved implementing and analyzing the Twin Rotor Multi-Input Multi-Output (TRMS) model using MATLAB and classical control techniques. The state-space model provided in Appendix A1 of the selected reference was used as the basis for all investigations.

Initially, the state-space matrices were implemented in MATLAB using the Control System Toolbox to construct the complete multi-input multi-output dynamic model.

The corresponding transfer function matrix was then derived, and the principal transfer functions were selected for detailed analysis.

Classical root locus techniques and proportional feedback control were employed to investigate the closed-loop behavior of the selected channels. Step response simulations and pole–zero maps were generated to evaluate transient performance, including rise time, settling time, peak time, and percentage overshoot.

To examine the structural properties of the TRMS model, open-loop eigenvalue analysis, controllability and observability tests, the PBH observability criterion, observability Gramian, Hankel singular values, minimal realization, and null-space analysis were performed. These methods were used to investigate system stability and identify hidden dynamic modes.

Finally, the results obtained from the classical control and state-space analyses were interpreted to provide a comprehensive evaluation of the dynamic and structural characteristics of the TRMS system.

4 Mathematical Modeling of the TRMS System

The Twin Rotor Multi-Input Multi-Output (TRMS) system is a nonlinear, highly coupled dynamic system commonly used as a benchmark for control system analysis and controller design. In this study, the state-space model provided in AppendixA1 of the selected reference was adopted as the mathematical representation of the TRMS and implemented in MATLAB for further analysis.

The general state-space representation of a linear dynamic system is given by

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{1}$$

and

$$y(t) = Cx(t) + Du(t), \tag{2}$$

where $x(t)$, $u(t)$, and $y(t)$ denote the state, input, and output vectors, respectively, while A, B, C, and D represent the state, input, output, and feedthrough matrices.

The implemented TRMS model consists of six state variables, two control inputs, and two measured outputs, providing a complete mathematical description of the system dynamics.

Table 1: Summary of the TRMS State-Space Model

Parameter	Value
System Order	6
Number of Inputs	2
Number of Outputs	2
Number of States	6
System Representation	State Space
Source	Appendix A1

The complete state-space matrices used in this investigation are provided in AppendixA. These matrices were implemented directly in MATLAB and served as the basis for transfer function derivation and subsequent analyses.

The state-space model forms the mathematical foundation of this study, enabling transfer function derivation, classical root locus analysis, transient response evaluation, stability assessment, controllability and observability investigations, and reduced-order system analysis. Consequently, it provides a comprehensive framework for examining both the dynamic and structural characteristics of the TRMS system.

5 Transfer Function Derivation and Analysis

Transfer functions describe the input–output behavior of a dynamic system in the Laplace domain and provide the basis for classical control analysis. For a linear time-invariant system represented in state-space form, the transfer function matrix is given by

$$G(s) = C(sI - A)^{-1}B + D, \tag{3}$$

where A, B, C, and D are the state-space matrices and I is the identity matrix. In this study, the transfer function matrix was obtained from the implemented TRMS state-space model using the MATLAB Control System Toolbox.

The TRMS is a two-input, two-output dynamic system, resulting in a transfer function matrix with four input–output channels. Analysis of the obtained model showed that two channels contain the dominant observable dynamics and were therefore selected for detailed classical control investigations.

The selected transfer functions are

$$G_1(s) = \frac{1.359}{s^3 + 0.9973s^2 + 4.786s + 4.278}, \tag{4}$$

and

$$G_2(s) = \frac{4.5}{s^3 + 2.526s^2 + 6.47s + 4.545}. \tag{5}$$

Both transfer functions are third-order systems representing the principal dynamic modes of the TRMS model. These channels were selected for further analysis because they capture the significant input–output behavior of the system.

The derived transfer functions establish the link between the state-space representation and classical control methods and serve as the basis for the root locus, transient response, and closed-loop analyses presented in the following section.

6 Classical Root Locus and Closed–Loop Analysis

6.1 Analysis of the First Transfer Function

The first principal transfer function selected from the TRMS model is

$$G_1(s) = \frac{1.359}{s^3 + 0.9973s^2 + 4.786s + 4.278}. \tag{6}$$

The transfer function represents one of the principal dynamic modes of the TRMS system. Root locus analysis and proportional feedback control were employed to investigate its stability and transient response characteristics.

The root locus shown in Figure 1 illustrates the movement of the closed-loop poles as the proportional gain varies. The pole trajectories remain within the stable region, indicating stable closed-loop operation over the selected gain range.

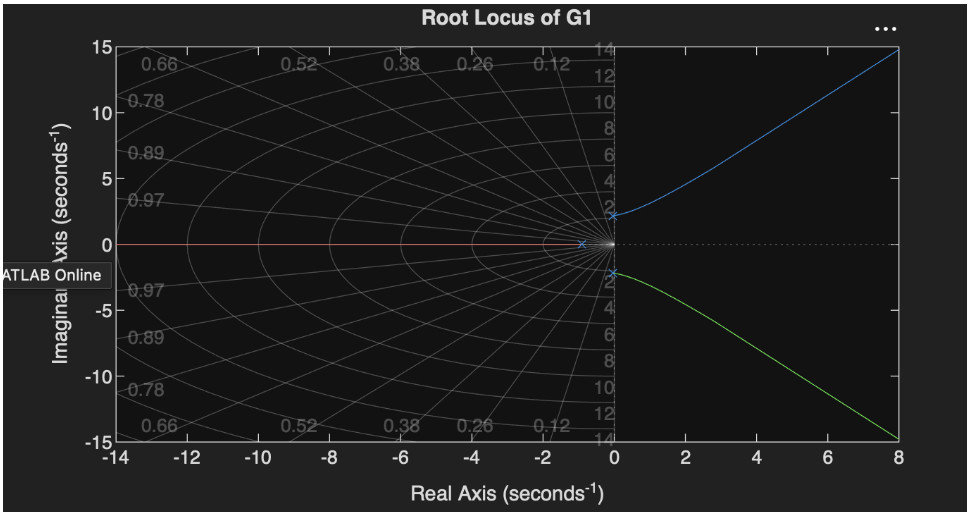

Figure 1: Root Locus of G1

The corresponding closed-loop step response is presented in Figure 2. The response was used to evaluate the transient performance of the system, including rise time, settling time, peak time, and overshoot.

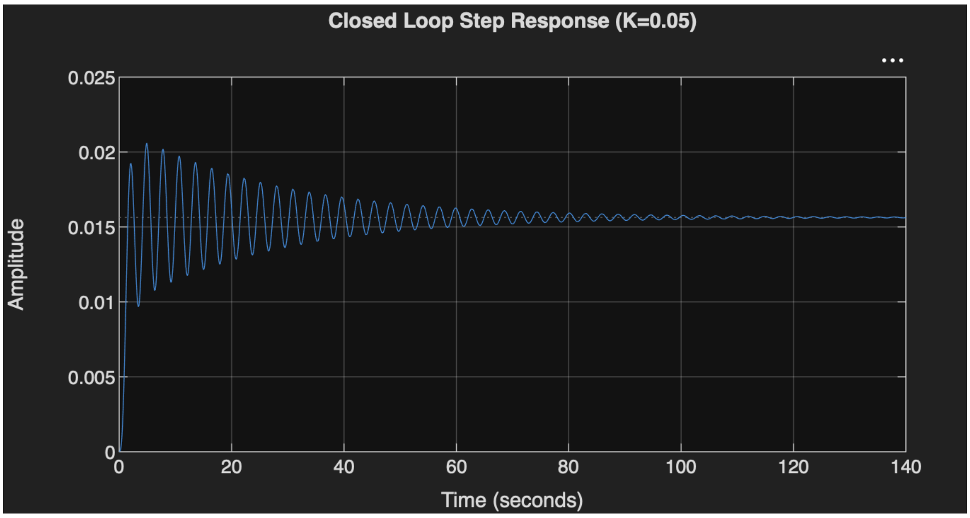

Figure 2: G1 Step Response

The step response exhibits a rapid initial rise followed by moderate oscillations before reaching steady state. The associated pole–zero map is shown in Figure 3.

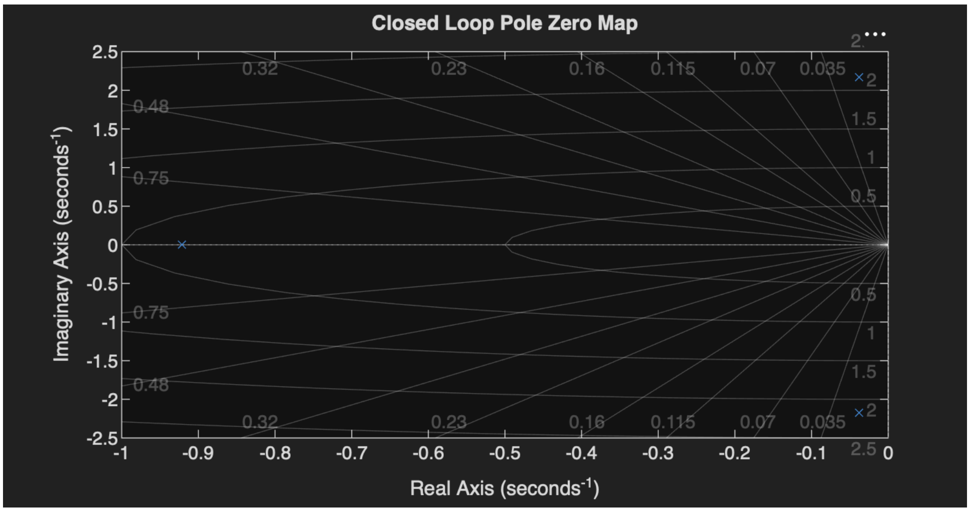

Figure 3: G1 Pole-Zero Map

The pole–zero map confirms that the closed-loop poles lie in the left-half of the complex plane, verifying the stable behavior predicted by the root locus and transient response analyses.

6.2 Analysis of the Second Transfer Function

The second principal transfer function selected from the TRMS model is

$$G_2(s) = \frac{4.5}{s^3 + 2.526s^2 + 6.47s + 4.545}. \tag{7}$$

Similar to $G_1(s)$, the second transfer function was investigated using root locus analysis and proportional feedback control.

The root locus shown in Figure 4 illustrates the variation of the closed-loop poles with proportional gain. The pole trajectories remain within the stable region, indicating stable closed-loop performance.

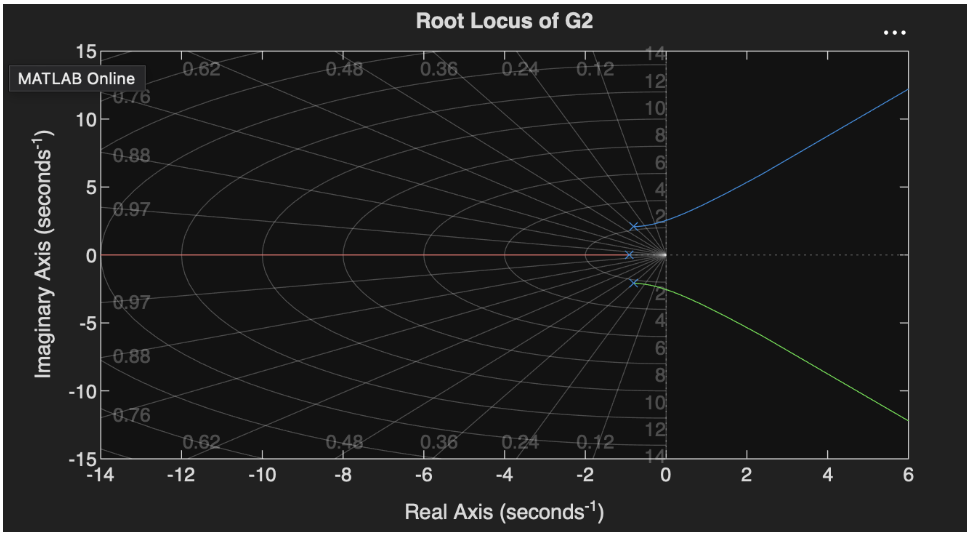

Figure 4: G2 Root Locus

The closed-loop step response obtained for the selected gain is presented in Figure 5. The response was used to evaluate the transient characteristics of the system under proportional feedback control. The corresponding pole–zero map is shown in Figure 6.

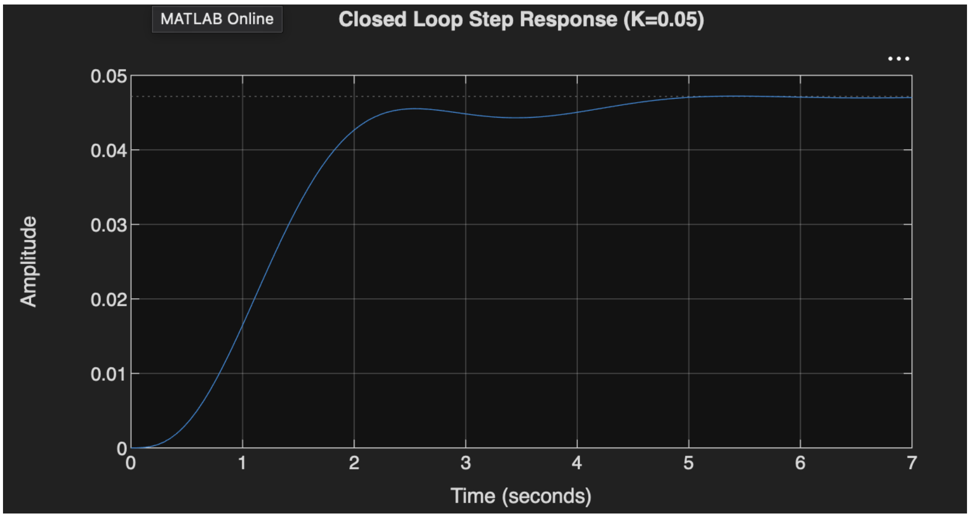

Figure 5: Closed Loop Step Response

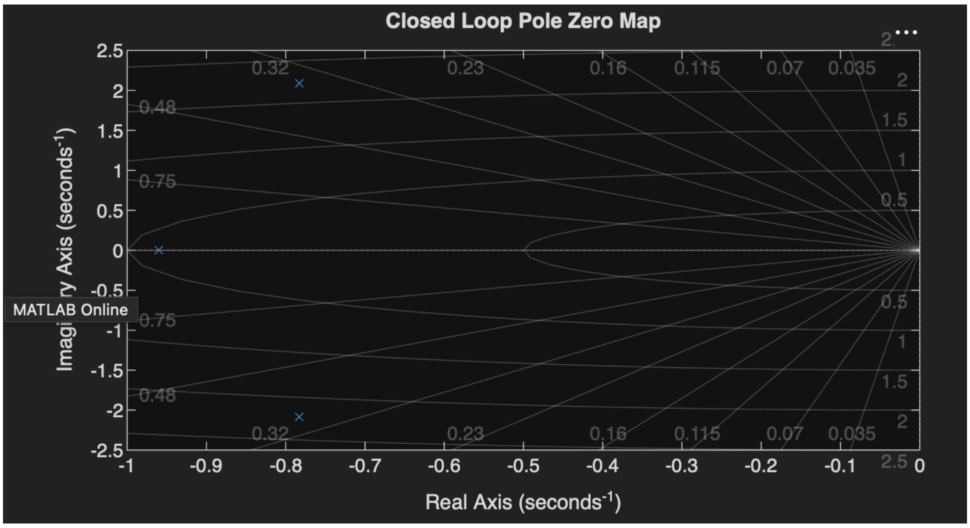

Figure 6: G2 Pole-Zero Map

The pole–zero distribution confirms that the closed-loop poles remain in the left-half of the complex plane, consistent with the stable behavior observed from the root locus and step response analyses.

The classical investigations of $G_1(s)$ and $G_2(s)$ demonstrate that both principal channels exhibit stable closed-loop operation under proportional feedback control. These results provide the basis for the structural state-space investigations presented in the following section.

7 Open–Loop Structural Analysis

7.1 Stability

The open-loop stability of the TRMS model was investigated by examining the eigenvalues of the state matrix. For a continuous-time linear system, stability is achieved when all eigenvalues possess negative real parts.

The eigenvalue distribution obtained from the MATLAB implementation is shown in Figure 7.

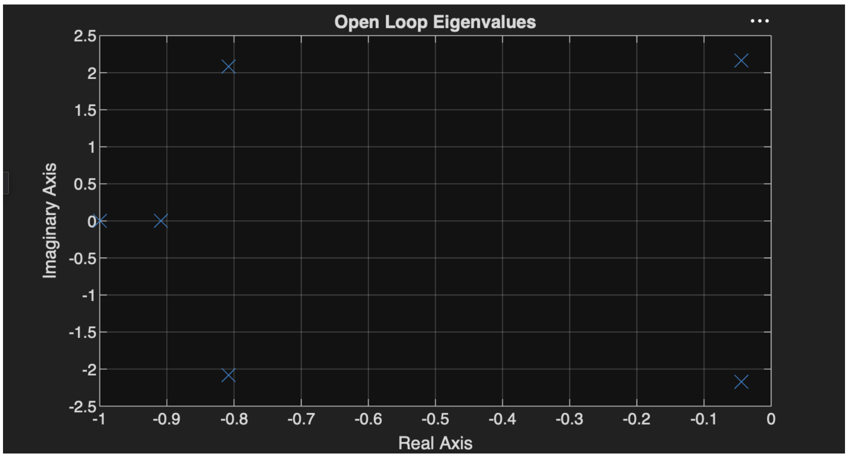

Figure 7: pen-loop eigenvalue distribution of the TRMS state-space model.

The open-loop eigenvalues are

$$\begin{aligned}
\lambda_1 &= -0.0441 + 2.1689i, \\
\lambda_2 &= -0.0441 - 2.1689i, \\
\lambda_3 &= -0.8085 + 2.0848i, \\
\lambda_4 &= -0.8085 - 2.0848i, \\
\lambda_5 &= -0.9091, \\
\lambda_6 &= -1.0000.
\end{aligned} \tag{8}$$

All eigenvalues lie within the left-half of the complex plane, confirming that the TRMS model is inherently stable under open-loop conditions. The complex conjugate pairs correspond to oscillatory modes, while the negative real eigenvalues represent exponentially decaying responses. This result provides the basis for the subsequent structural analyses.

7.2 Controllability Analysis

Controllability describes the ability to drive a system from any initial state to any desired final state through suitable control inputs. For an nth-order system, the controllability matrix is

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}. \tag{9}$$

The controllability matrix of the TRMS model was constructed in MATLAB and its rank was evaluated. The analysis yielded

$$\text{Rank}(\mathcal{C}) = 6. \tag{10}$$

Since the rank is equal to the order of the system, all six state variables are controllable. This indicates that every dynamic mode of the TRMS model can be influenced through the available control inputs.

The controllability analysis confirms that the implemented TRMS model is completely controllable and suitable for feedback control design, providing a foundation for the observability investigations presented in the following sections.

7.3 Observability Analysis

Observability describes the ability to reconstruct the internal states of a system from its measured outputs. For a linear time-invariant system, the observability matrix is defined as

$$\mathcal{O} = \left[C \; CA \; CA^2 \; \vdots \; CA^{n-1} \right], \tag{11}$$

where A and C are the state and output matrices.

The observability matrix of the TRMS model was constructed in MATLAB, and its rank was evaluated to determine whether all state variables contribute to the measured outputs. The analysis yielded

$$\text{Rank}(\mathcal{O}) = 5. \tag{12}$$

Since the TRMS model is sixth order, complete observability would require a rank of six. The obtained result indicates that one state variable cannot be reconstructed from the available output measurements.

The observability analysis therefore demonstrates that the TRMS model is not completely observable and contains one hidden dynamic mode. This observation motivates the subsequent structural investigations used to identify and characterize the unobservable state.

7.4 PBH Observability Test

To identify the hidden dynamic mode, the Popov–Belevitch–Hautus (PBH) observability criterion was applied. For a linear time-invariant system, the PBH test requires

$$\text{Rank} \left[\lambda I - A \; C \right] = n, \tag{13}$$

for every eigenvalue λ of the state matrix.

The PBH analysis was performed in MATLAB by evaluating the rank of the augmented matrix associated with each eigenvalue of the TRMS model. Five eigenvalues satisfied the observability condition, while one eigenvalue produced a rank deficiency.

The unobservable mode was identified as

$$\lambda = -1. \tag{14}$$

For this eigenvalue, the augmented matrix had a rank of five rather than six, indicating that the corresponding dynamic mode cannot be reconstructed from the measured outputs.

The PBH test confirms the results obtained from the observability matrix and identifies the exact mode responsible for the loss of complete observability. The presence of a single unobservable but internally stable mode motivates the subsequent Gramian, Hankel singular value, and minimal realization analyses.

7.5 Observability Gramian Analysis

The observability Gramian provides a measure of the contribution of the internal system states to the measured outputs. For a stable continuous-time system, the observability Gramian W_o satisfies

$$A^T W_o + W_o A = -C^T C, \tag{15}$$

where A and C are the state and output matrices.

The observability Gramian of the TRMS model was computed in MATLAB, and the corresponding eigenvalues were obtained as

$$0, ; 0.0175, ; 0.4513, ; 0.4967, ; 1.3289, ; 6.0328. \tag{16}$$

The presence of a zero eigenvalue indicates that one dynamic mode does not contribute to the measured outputs and is therefore unobservable, while the remaining positive eigenvalues correspond to observable system modes.

The observability Gramian independently confirms the conclusions obtained from the observability matrix and the PBH observability criterion, providing further evidence for the existence of a single hidden dynamic mode within the TRMS model.

7.6 Hankel Singular Value Analysis

Hankel singular values combine controllability and observability information and provide a measure of the relative importance of the system states. They are commonly used in structural analysis and model reduction.

The Hankel singular values of the TRMS model were computed in MATLAB and are given by

$$1.5610, ; 1.5156, ; 0.6587, ; 0.2579, ; 0.0864, ; 0. \tag{17}$$

The results indicate that five dynamic modes contribute to the input–output behavior of the system, while the sixth Hankel singular value is zero, indicating the presence of a redundant dynamic mode.

The Hankel singular value analysis confirms the results obtained from the observability matrix, PBH test, and observability Gramian. The existence of a zero singular value suggests that one state can be removed without affecting the external input–output characteristics of the TRMS model, motivating the minimal realization analysis presented in the following section.

7.7 Minimal Realization

Minimal realization is a model reduction technique that removes uncontrollable or unobservable states while preserving the external input–output behavior of a system.

The minimal realization of the TRMS model was obtained using the MATLAB Control System Toolbox. The analysis showed that the original sixth-order state-space model can be reduced to a fifth-order realization without altering its transfer function characteristics.

The reduction in system order is consistent with the results obtained from the observability matrix, the PBH observability criterion, the observability Gramian, and the Hankel singular value analysis. Each of these methods indicated the presence of a single unobservable dynamic mode, and the minimal realization confirms that this mode can be removed without affecting the observable system behavior.

The reduced-order model preserves the essential dynamic characteristics of the TRMS system while providing a simpler mathematical representation for analysis and control design. This result motivates the final structural investigation, where the hidden dynamic mode is identified through null-space analysis.

7.8 Hidden State Analysis

The previous structural analyses established that the TRMS model contains a single unobservable dynamic mode. To identify the corresponding state variable, the null space of the observability matrix was evaluated in MATLAB.

The null-space analysis yielded

$$N = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{18}$$

The result indicates that the sixth state variable, x_6, is unobservable and cannot be reconstructed from the available output measurements.

This finding is consistent with the observability matrix, the PBH observability criterion, the observability Gramian, the Hankel singular value analysis, and the minimal realization of the TRMS model. All structural investigations independently identified the existence of a single hidden dynamic mode.

Although the sixth state remains internally stable, it does not influence the observable input–output behavior of the system. Consequently, its removal during the minimal realization process does not alter the external dynamics of the TRMS model.

The hidden state analysis completes the structural investigation of the TRMS system. The combined results demonstrate that the model is open-loop stable, completely controllable, but not completely observable due to the presence of one internally stable and unobservable state. The resulting fifth-order realization therefore provides an equivalent and more compact representation of the system dynamics.

8 Discussion

The analyses performed in this study provide a comprehensive understanding of both the dynamic behavior and structural characteristics of the TRMS model. The derived transfer functions established the connection between the state-space representation and classical control analysis, while root locus, step response, and pole–zero investigations demonstrated stable closed-loop operation under proportional feedback control.

The state-space investigations provided additional insight into the internal properties of the system. Open-loop eigenvalue analysis confirmed the inherent stability of the TRMS model, and controllability analysis showed that all dynamic states can be influenced through the available control inputs. In contrast, observability analysis revealed the presence of a single hidden dynamic mode.

The observability matrix, PBH observability criterion, observability Gramian, Hankel singular value analysis, minimal realization, and null-space investigation consistently identified one internally stable but unobservable state. The minimal realization further demonstrated that the original sixth-order model can be reduced to a fifth-order representation without altering its observable input–output behavior.

The combined results show that the TRMS model possesses desirable stability and controllability characteristics while containing one redundant dynamic mode. Overall, the integration of classical control and state-space methods provides a comprehensive framework for understanding the dynamic and structural behavior of complex multi-input multi-output systems.

Appendix A: TRMS State–Space Matrices

The complete state–space matrices employed throughout this investigation are presented in this appendix. These matrices were adopted from Appendix A1 of the selected reference and implemented directly in MATLAB for transfer function derivation, classical root locus analysis, and structural system investigations. Together, they provide the mathematical foundation for all analyses performed in this study.

The TRMS model considered in this work is a sixth-order, two-input, two-output dynamic system represented in the standard state-space form

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{19}$$

and

$$y(t) = Cx(t) + Du(t), \tag{20}$$

where A, B, C, and D denote the state, input, output, and feedthrough matrices, respectively.

A State Matrix

The state matrix describes the internal dynamic relationships among the six state variables of the TRMS model.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -4.7059 & -0.0882 & 0 & 0 & 1.3588 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & -1.617 & 4.5 & 0 \\ 0 & 0 & 0 & 0 & -0.9091 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \tag{21}$$

B Input Matrix

The input matrix defines the influence of the two control inputs on the system state variables.

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0.8 \end{bmatrix} \tag{22}$$

C Output Matrix

The output matrix specifies the measured outputs of the TRMS system.

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \tag{23}$$

D Feedthrough Matrix

The feedthrough matrix represents the direct relationship between the system inputs and outputs.

$$D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{24}$$

The matrices presented in this appendix define the complete mathematical model of the TRMS system and form the basis for all transfer function, classical control, and state-space analyses performed throughout this investigation.

Appendix B: MATLAB Implementation

This appendix presents the MATLAB programs developed for the implementation and analysis of the TRMS model. The scripts were used for mathematical modeling, transfer function derivation, classical control investigations, and structural system analyses. The implementations provided in this appendix allow the principal results presented throughout this report to be reproduced.

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E State–Space Model Implementation

The following MATLAB script was used to implement the sixth-order TRMS state-space model employed throughout this investigation.

```

clc;
clear;
close all;

A = [
0        1        0        0        0        0;
-4.7059  -0.0882  0        0        1.3588   0;
0        0        0        1        0        0;
0        0       -5       -1.617    4.5      0;
0        0        0        0       -0.9091   0;
0        0        0        0        0       -1
];

B = [
0 0;
0 0;
0 0;
0 0;
1 0;
0 0.8
];

```

```
C = [
1 0 0 0 0 0;
0 0 1 0 0 0
];

D = [
0 0;
0 0
];

sys = ss(A,B,C,D);

disp(sys)
```